

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, ML Systems & Evaluation Aug 2025 to May 2026

- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Proved a **2.4 pt** ensemble gain was real with **McNemar** testing at $p < 0.001$ instead of reporting a single accuracy number, across **1,700** held-out samples.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.

Turito | Software Engineering Intern, Systems Performance Jun 2022 to Oct 2022

- Slashed p95 CPU **22%** by profiling hot paths and removing per-packet allocations, payload copies, redundant lookups, and **lock contention** in shared state.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Prototyped a **Rust** + **QUIC** media path serving about **8,000** lightweight subscribers with about **24%** lower startup time.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: C++17, Python, Rust, C, SQL

Coding Agents & Tooling: Claude Code, OpenAI Codex, agent evaluation, failure classification, targeted test generation, diff review, garak

Performance & Low Level: AVX2/SIMD kernels, int8 quantization, cache-aware memory layout, profiling, multithreading, CMake, pybind11, Linux

ML: PyTorch, ViT, ResNet, ensembles, Hugging Face, GPU programming coursework

PROJECTS

Teller, Agent Evaluation Environment | *Browser agents, Claude Code, Python, TypeScript*

- Analyzed **14** computer-use agent runs against an adversarial app (nested frames, missing test IDs, session expiry), classifying each failure as planning, execution, validation, or recovery.
- Ported the same workflows to a second tenant with a **16-line** overlay and zero re-recording by designing a reusable capability layer.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.
- Cut traversal memory **75%** (51 MB to 12.8 MB) with **int8** AVX2 traversal plus exact float re-ranking and cache-aware node reordering.

OPEN SOURCE

- NVIDIA/garak:** preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- fission/fission:** 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: GPU Programming, Advanced Algorithms, Machine Learning, Distributed Computing, Artificial Intelligence, Database Systems

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)